

THOMAS MEBARKI

Lille, France | +33 6 22 51 59 34 | thomas.mebarki@protonmail.com | linkedin.com/in/thomasmebarki | github.com/ThomasMeb | mebarki.dev | egir.app

PROFILE

ML Engineer who left tech to work as a cook — and came back with a SaaS. I build models that serve the business, not the other way around.

3 years of applied Machine Learning experience (Fraud Detection, Time Series, NLP, Computer Vision, Segmentation), with a product-focused, ROI-driven approach. Atypical background: field immersion in restaurants to understand business constraints before creating EGIR, an analytics SaaS for restaurant owners in commercial launch.

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, Clustering (K-Means, PCA), Anomaly Detection (Isolation Forest, Autoencoders)

Deep Learning: TensorFlow, PyTorch, Keras, CNN, Transfer Learning (ResNet50V2, VGG16, EfficientNet)

NLP & LLMs: Transformers, BERT, Mistral, RAG, Hugging Face, LangChain / LangGraph, TF-IDF, Embeddings, Multi-label Classification

MLOps & Infra: Docker, Docker Compose, GitHub Actions CI/CD, Caddy, OVH VPS, MLflow, SHAP, pytest, Ruff, mypy

Data Engineering: Python 3.12, SQL, PostgreSQL, Pandas, NumPy, PySpark, Databricks, Feature Engineering

Full-Stack: FastAPI, Next.js, React, Streamlit, SQLAlchemy, Stripe, Supabase, pgvector

Cloud & Tools: AWS, GCP, Vercel, Cloudflare (DNS + R2), Linux, Jupyter, VS Code

PROFESSIONAL EXPERIENCE

Founder & ML Engineer — EGIR

August 2025 - Present

Lille | egir.app

Analytics SaaS for restaurant owners, in commercial launch phase (B2G pilot with Mairie de Gravelines). Solo-built from product design to deployment.

- Designed and developed an end-to-end SaaS product: FastAPI backend, Next.js 16 frontend, multi-tenant, RBAC, freemium
- Integrated a conversational AI assistant (Mistral LLM) for guided recipe creation — 10+ sequential steps, multi-turn conversation
- Implemented the cost propagation engine: recursive hierarchy with automatic recalculation
- Deployed sovereign infrastructure on OVH VPS: Docker Compose, Caddy, daily Cloudflare R2 backups, CI/CD GitHub Actions
- Integrated Stripe (Checkout, Portal, webhook) and freemium model with feature gates (EUR 79 launch / EUR 119 regular)
- Complete admin panel (6 sprints): revenue MRR tracking, analytics, support inbox, collectives management

Key skills: Python, FastAPI, Next.js, PostgreSQL, Mistral LLM, Docker, OVH, Stripe, Product Management

ML Engineer — Supervised Fraud Detection (Internship)

January 2023 - June 2023

Shift Technology (Insurtech, French unicorn) — Paris

Claims scoring for SIU (Special Investigation Unit) teams.

- Explored multi-insurer transactional data at scale via PySpark/Databricks
- Built transactional and temporal features (underwriting-to-claim delay, rolling windows, contract age)
- Developed supervised scoring models (XGBoost/LightGBM) on labeled data

- Handled class imbalance (fraud 1-3%) via `scale_pos_weight` and Precision-Recall AUC optimization

- Evaluated via Precision@K (SIU metric), decile-wise Lift, hit rate. SHAP values for interpretability

Key skills: Python, PySpark, Databricks, XGBoost, SHAP, Feature Engineering, Statistical Testing

ML Engineer — Behavioral Anomaly Detection (Freelance)

July 2023 – November 2023

Shift Technology (Insurtech) — Paris

Freelance extension. Detecting novel fraud patterns not captured by supervised models.

- Implemented unsupervised anomaly detection: Isolation Forest, autoencoders
- Built social graph features (policyholders, garages, experts) to detect fraud rings
- Set up production drift monitoring: PSI (Population Stability Index) and KS tests
- Measured novel fraud discovery rate (frauds caught by unsupervised but missed by supervised)

Key skills: Isolation Forest, Autoencoders, NetworkX, PSI, MLOps Monitoring

Data Scientist — Energy Prediction

November 2023 – December 2023

Schneider Electric

- Predictive model for building energy consumption: $R^2 = 0.83$, +45% vs linear baseline
- Temporal feature engineering (sin/cos encoding, lag J-1, J-7) and weather features (HDD / CDD)
- XGBoost benchmarked against Random Forest and LightGBM, GridSearchCV tuning
- SHAP values for interpretability (HDD, CDD, surface, lag J-1 dominant)

Key skills: XGBoost, Temporal Feature Engineering, HDD/CDD, SHAP, Regression, Time Series

Data Scientist — Customer Segmentation

December 2023 – February 2024

Back Market (Refurbished E-commerce)

- RFM segmentation on ~750,000 customers (France scope, 24-month transactional history)
- K-Means clustering (K=4, Silhouette = 0.49) validated via Elbow method and PCA
- 4 actionable segments: VIP (5% base, 30% revenue), Regulars, Occasionals, One-shot
- +25% email open rate after per-segment personalization

Key skills: K-Means, PCA, RFM Analysis, Silhouette Score, Python, Data Visualization

NLP Engineer — Tag Recommendation System

March 2024 – May 2024

Stack Overflow

- Multi-label NLP classification (sigmoid per tag + Binary Cross-Entropy)
- Multi-model comparison: TF-IDF (~65%), Word2Vec (~68%), USE (~72%), BERT fine-tuned selected (78%)
- Precision@5 = 78%, Recall@5 = 62%, p95 latency = 145ms
- 31% reduction in moderator corrections, 52% user adoption

Key skills: BERT, Transformers, TF-IDF, Word2Vec, FastAPI, Multi-label Classification

ML Engineer — Computer Vision

June 2024 – September 2024

SantéVet (Insurtech, French leader in pet insurance)

- Image classifier for 120 dog breeds (Stanford Dogs Dataset enriched)
- 3 CNN backbone comparison: VGG16, EfficientNet-B3, ResNet50V2 selected (Top-3 Accuracy = 90%)
- Original architecture: CNN extractor + RBF SVM classifier + Platt scaling (reliable probabilities for pricing)
- <500ms latency on CPU, FastAPI API in Docker container

Key skills: CNN, Transfer Learning, ResNet50V2, RBF SVM, TensorFlow, Keras, OpenCV

Business Immersion — Artisan Cook

December 2024 – July 2025

Café de Paris, Calais

Voluntary 8-month immersion in a professional kitchen to understand operational constraints of restaurants from the inside.

- Actual kitchen work: mise en place, service, rush management (lunch + dinner, split shift)

- Parallel training: Artisan Cook Diploma at a Master Restaurateur
- Cost structure mapping, profitability lever identification, waste quantification
- Transformed observations into product specifications for EGIR

Key skills: Business Analysis, Domain Understanding, Field Observation

PERSONAL PROJECTS

JobScout (github.com/ThomasMeb/JobScout): Autonomous job search SaaS. 9 scrapers, 3-tier scoring (pgvector + DeepSeek LLM), Notion sync, 140+ tests. Next.js + FastAPI + Supabase, deployed on OVH.

Grada: On-chain BTC trading bot via dHEDGE (Polygon). 61% accuracy, Sharpe 3.37. Python (ML) + TypeScript (blockchain). Currently in DRY_RUN post-audit.

Portfolio mebarki.dev: Next.js showcase with interactive ML demos for the 7 ML/DS projects.

EDUCATION

Master's in Data Science — Machine Learning specialization | *CentraleSupélec* | 2023

Bachelor's in Mathematics — With distinction | *Université du Littoral Côte d'Opale (ULCO)* | 2021

Artisan Cook Diploma | *À l'École des Chefs (Master Restaurateur)* | 2025

CERTIFICATIONS & LANGUAGES

CLES B2 English · PSC1 first aid · French (native) · English (professional, 4/5)

ADDITIONAL INFORMATION

Karate black belt, departmental champion · Professional Muay Thai fight won in Chiang Mai (Thailand) · Marathon runner, sailor (competitive regattas) · Teaching experience: private math tutor, university L3 tutor